

Privacy-Preserving Machine Learning with Federated Learning and Differential Privacy: A Survey

Privacy-preserving machine learning studies how models can be trained, evaluated, and deployed while limiting the exposure of sensitive data. Federated learning reduces raw-data centralization by moving optimization to clients or organizations, whereas differential privacy provides formal limits on what an adversary can infer from released models, gradients, or statistics. This survey synthesizes an arXiv-derived corpus of 300 papers collected during this run and organizes the literature around federated optimization, differential privacy theory and accounting, DP-SGD, secure aggregation, cryptographic mechanisms, privacy attacks, robustness under heterogeneous data, application domains, and evaluation methodology. The survey argues that privacy-preserving ML is an end-to-end systems problem: formal privacy, statistical utility, adversarial leakage, communication, deployment constraints, and governance must be designed together.

Keywords:

privacy-preserving machine learning; federated learning; differential privacy; DP-SGD; secure aggregation; privacy attacks; private deep learning

1 Introduction

Machine learning increasingly relies on sensitive data from phones, hospitals, financial institutions, homes, vehicles, and online platforms. The resulting models can improve prediction and personalization, but they can also leak information through training records, gradients, embeddings, model outputs, or deployment logs. Privacy-preserving machine learning therefore aims to make useful learning compatible with data minimization, formal privacy, and organizational constraints [103, 104, 143, 147, 180, 181, 201, 286].

Federated learning and differential privacy are two central pillars. Federated learning keeps raw data on clients or within institutional silos while sharing model updates. Differential privacy provides a mathematical guarantee that the presence or absence of an individual record has limited influence on released outputs. In practice, these ideas are often combined with secure aggregation, cryptographic computation, robust optimization, personalization, and privacy auditing [104, 143, 147, 167, 180, 181, 185, 286].

This survey reviews privacy-preserving ML through the interaction between algorithms, systems, threat models, and deployment settings. We focus on how FL and DP complement each other, where they leave gaps, and how evaluation should measure both privacy and utility.

2 Background

2.1 Federated Learning

Federated learning trains a model by repeatedly selecting clients, sending model state, collecting local updates, and aggregating those updates on a server. Cross-device FL involves many unreliable clients with limited compute and intermittent connectivity; cross-silo FL involves fewer institutions with higher availability but stronger governance requirements. Federated optimization must handle non-IID data, stragglers, communication cost, client drift, and statistical heterogeneity [104, 143, 147, 167, 180, 181, 185, 286].

2.2 Differential Privacy

Differential privacy bounds how much an output distribution can change when one individual’s record changes. Central DP assumes a trusted curator that adds noise before release, while local DP randomizes data or updates before they leave the client. Privacy accounting tracks cumulative privacy loss across many training steps, often using Renyi or concentrated variants. The central tension is that stronger privacy usually requires more noise, which can reduce utility [103, 143, 166, 180, 181, 185, 201, 286].

2.3 Private Deep Learning and DP-SGD

DP-SGD clips per-example gradients and adds calibrated noise before updating model parameters. It is widely used because it integrates with deep learning, but it introduces practical challenges: per-example gradient computation is expensive, clipping interacts with optimization, privacy budgets accumulate, and large models can be sensitive to noise. Private training therefore requires careful choices of batch size, clipping norm, learning rate, pretraining, fine-tuning, and accounting [103, 104, 166, 180, 181, 201, 248, 300].

2.4 Secure Aggregation and Cryptographic Mechanisms

Secure aggregation hides individual client updates from the server while allowing aggregate computation. It is often combined with FL to reduce server-side visibility into gradients. Homomorphic encryption, secure multi-party computation, secret sharing, and trusted execution environments provide additional options, each with different assumptions and overheads. Cryptographic privacy and DP are complementary: cryptography can hide updates during computation, while DP limits what the final output reveals [97, 143, 147, 160, 170, 185, 251, 286].

3 Methods

3.1 Search Protocol and Corpus Construction

The corpus for this survey was generated during this run from the arXiv structured API. Queries combined the task topic with privacy-preserving machine learning, federated learning, differential privacy, differentially private FL, federated optimization, FedAvg, FedProx, cross-device and cross-silo FL, non-IID data, personalized and fair FL, DP-SGD, private SGD, gradient clipping, privacy accounting, Renyi

DP, privacy amplification, local DP, secure aggregation, secure multi-party computation, homomorphic encryption, trusted execution, membership inference, model inversion, gradient inversion, gradient leakage, data reconstruction, privacy utility trade-offs, healthcare, mobile, edge, IoT, foundation models, and benchmarks. Records were retained when title or abstract evidence connected machine learning to FL, DP, private training, secure aggregation, privacy attacks, privacy leakage, privacy-preserving deployment, or evaluation. Duplicates were removed by arXiv identifier, and the final bibliography cites 300 unique scholarly preprints.

3.2 Taxonomy

We organize the literature along eight axes. The first is *learning topology*: centralized, local, cross-device federated, cross-silo federated, decentralized, or split. The second is *privacy mechanism*: DP noise, local randomization, secure aggregation, cryptography, trusted execution, or access control. The third is *threat model*: honest-but-curious servers, malicious clients, colluding participants, external attackers, or model consumers. The fourth is *leakage channel*: data, gradients, updates, embeddings, model outputs, or logs. The fifth is *utility objective*: accuracy, calibration, personalization, fairness, robustness, or downstream task quality. The sixth is *system constraint*: communication, device availability, computation, memory, energy, and latency. The seventh is *domain*: healthcare, mobile, IoT, finance, recommender systems, language models, and edge AI. The eighth is *governance*: consent, deletion, auditability, accountability, and legal compliance.

3.3 Federated Optimization Under Privacy Constraints

Privacy changes the optimization problem. Secure aggregation can limit the server’s visibility into updates, but it can also constrain debugging, anomaly detection, and personalization. Differential privacy injects noise into updates or model parameters, which can amplify the effect of client heterogeneity and small client samples. Practical systems must choose client sampling, clipping, aggregation, compression, and personalization strategies jointly [104, 143, 147, 167, 180, 181, 185, 286].

3.4 Differential Privacy Accounting and Amplification

Privacy accounting translates training choices into formal guarantees. Composition accumulates pri-

vacy loss over many rounds; subsampling can amplify privacy; adaptive training schedules complicate accounting. For FL, accounting must consider client sampling, example sampling within clients, secure aggregation, dropout, and whether guarantees are at the record, user, device, or institution level [103, 143, 166, 180, 181, 185, 201, 286].

3.5 Private Training of Deep and Foundation Models

DP-SGD has moved from small classifiers to large neural networks and foundation-model fine-tuning. Transfer learning, public pretraining, parameter-efficient adaptation, and careful clipping can improve the privacy-utility trade-off. Yet large models create new risks: memorization, prompt leakage, membership inference, and auditing difficulty. Private foundation-model training therefore requires both formal DP and empirical privacy evaluation [103, 104, 166, 180, 181, 201, 248, 300].

3.6 Secure Aggregation and System Design

Secure aggregation protocols must tolerate dropped clients, scale to many participants, and integrate with compression, clipping, and DP noise. Cross-silo deployments may prefer heavier cryptographic protocols because participants are more stable, while cross-device deployments require lightweight protocols and robust dropout handling. System design therefore determines whether a theoretically private method remains usable in realistic deployments [97, 143, 147, 160, 170, 185, 251, 286].

3.7 Privacy Attacks and Auditing

Attacks reveal what privacy mechanisms must defend against. Membership inference tests whether a record influenced a model. Model inversion and reconstruction attempt to infer input data. Gradient leakage can reconstruct examples from shared updates, especially in small-batch or overparameterized settings. Auditing uses attacks, canaries, and empirical privacy estimates to test whether claimed protections hold under realistic adversaries [103, 147, 166, 167, 180, 181, 201, 286].

3.8 Robustness, Personalization, and Fairness

FL and DP interact strongly with heterogeneity. Non-IID data can cause client drift and unequal

utility. Personalization can improve local performance but may weaken shared privacy accounting or reveal client-specific patterns. Robustness methods mitigate poisoning and Byzantine behavior, but they may require visibility into updates that privacy mechanisms try to hide. Fair private learning must track utility disparities across clients and groups [104, 118, 121, 143, 181, 185, 286, 300].

3.9 Applications

Healthcare, mobile keyboards, recommender systems, IoT, wireless networks, finance, and edge AI motivate much of the field. These settings differ in data sensitivity, regulation, client availability, label quality, and tolerance for accuracy loss. Healthcare often emphasizes cross-silo governance and auditability; mobile deployments emphasize cross-device scale and communication; edge and IoT systems emphasize energy, latency, and unreliable connectivity [104, 147, 167, 180, 181, 185, 286, 300].

3.10 Evaluation Methodology

Evaluation should report utility, privacy budget, threat model, accounting method, attack performance, communication, client sampling, heterogeneity, and system overhead. Comparing methods without the same privacy accountant, clipping strategy, client population, or threat model can be misleading. Benchmarks should include both formal DP guarantees and empirical leakage tests, because a formal guarantee may be too weak for a particular deployment and an empirical defense may fail under a stronger attack [103, 104, 143, 147, 180, 181, 201, 286].

3.11 Representative Literature Map

The following map cites the retained corpus by theme, making the coverage explicit across federated systems, differential privacy, private deep learning, secure aggregation, attacks, personalization, applications, and evaluation.

Federated learning systems and optimization The federated learning systems and optimization cluster contains work that develops methods, systems, datasets, or analyses central to this survey [104, 143, 147, 160, 167, 180, 181, 185, 300]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [2, 33, 44, 74, 170, 207, 219, 225, 251]. Additional retained studies broaden the cluster with alternative

architectures, training signals, data sources, and deployment assumptions [5, 134, 135, 142, 163, 210, 231, 240, 243]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [8, 53, 76, 188, 193, 220, 239, 272, 279]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [3, 68, 136, 211, 212, 247, 257, 267, 288]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [52, 56, 99, 146, 172, 204, 218, 228, 276]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [10, 12, 69, 79, 108, 111, 132, 151, 214]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [37, 43, 73, 77, 98, 119, 205, 213, 262]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [29, 61, 80, 109, 140, 263–265, 285]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [6, 87, 93, 161, 182, 183, 190, 242, 268]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [14, 46, 110, 116, 165, 168, 186, 277, 280]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [7, 25, 90, 127, 144, 149, 192, 266, 271]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [115, 152, 175, 189, 200, 209, 230, 258, 289]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [9, 36, 58, 66, 75, 102, 158, 206, 273]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [1, 78, 86, 133, 173, 178, 287, 291, 292]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [54, 60, 101, 122, 215, 216, 226, 261, 290]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [19, 20, 28, 57, 64, 162, 174, 237, 250]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [4, 94, 129, 137, 164, 223, 254, 283].

Differential privacy theory and accounting The differential privacy theory and accounting cluster contains work that develops methods, systems, datasets, or analyses central to this survey [17, 18, 23, 70, 71, 83, 169, 177, 238]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [27, 39, 59, 63, 85, 184, 202, 221, 278]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [191, 274].

Private deep learning and dp-sgd The private deep learning and DP-SGD cluster contains work that develops methods, systems, datasets, or analyses central to this survey [40, 41, 67, 120, 145, 150, 179, 241, 249]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [30, 65, 82, 107, 126, 171, 235, 260, 297]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [11, 48, 84, 154, 156, 275].

Secure aggregation and cryptographic privacy The secure aggregation and cryptographic privacy cluster contains work that develops methods, systems, datasets, or analyses central to this survey [128, 138, 198, 227, 233, 253, 270].

Privacy attacks and leakage The privacy attacks and leakage cluster contains work that develops methods, systems, datasets, or analyses central to this survey [35, 47, 88, 96, 124, 201, 203, 269].

Robustness, personalization, and non-iid data The robustness, personalization, and non-IID data cluster contains work that develops methods, systems, datasets, or analyses central to this survey [100, 157, 234, 245, 295].

Applications in healthcare, mobile, and edge ai The applications in healthcare, mobile, and edge AI cluster contains work that develops methods, systems, datasets, or analyses central to this survey [22].

Benchmarks, evaluation, and deployment The benchmarks, evaluation, and deployment cluster contains work that develops methods, systems, datasets, or analyses central to this survey [34, 97, 103, 118, 121, 166, 236, 248, 286]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [32, 55, 62, 91, 130, 139, 176, 222, 259]. Additional

retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [26, 49, 72, 89, 195, 224, 232, 246, 256]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [16, 105, 114, 141, 153, 155, 194, 217, 229]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [24, 81, 112, 123, 131, 148, 159, 197, 244]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [13, 45, 95, 106, 187, 208, 282, 294, 299]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [31, 92, 117, 125, 196, 255, 281, 284, 298]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [15, 21, 38, 42, 50, 113, 252, 293, 296]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [51, 199].

4 Challenges and Prospects

4.1 Bridging Formal Guarantees and Operational Threat Models

Formal DP statements are valuable, but deployments need to map them to concrete user, device, and institutional risks. Future work should make threat models explicit and provide tools that translate privacy budgets into operational decisions understandable to engineers, auditors, and stakeholders.

4.2 Scaling Private Learning to Foundation Models

Foundation models increase both utility and leakage risk. Efficient private fine-tuning, parameter-efficient DP, private retrieval, private evaluation, and auditing for memorization will be central as organizations train and adapt large models on sensitive corpora.

4.3 Joint Design of FL, DP, and Secure Aggregation

FL, DP, and secure aggregation are often studied separately, yet real systems combine them. Joint design should decide where clipping occurs, where noise is

added, how aggregation hides updates, how clients are sampled, and how failures affect accounting.

4.4 Fairness Under Privacy and Heterogeneity

Privacy noise and client heterogeneity can create uneven accuracy losses. Future benchmarks should measure group-level and client-level utility, not only average accuracy, and should study how personalization and fairness constraints affect privacy accounting.

4.5 Auditable Privacy-Preserving ML Pipelines

Private learning systems need audit trails for data selection, client sampling, accounting, model release, attacks, deletion, and consent. Auditable pipelines will be especially important in healthcare, finance, enterprise AI, and regulated edge deployments.

5 Conclusion

Privacy-preserving machine learning combines algorithmic, statistical, cryptographic, and systems challenges. Federated learning reduces raw-data centralization, differential privacy limits inference from outputs, secure aggregation hides client updates, and privacy attacks reveal remaining leakage channels. This survey reviewed the literature across these layers and argued that successful private ML requires end-to-end co-design of formal guarantees, optimization, system architecture, empirical auditing, and application governance.

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